

“A new type of dementia—called seizures”: temporal lobe epilepsy presenting as episodic neuropsychiatric symptoms

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Abstract

Introduction: Older adults frequently present with episodic mood changes and fluctuating memory lapses, which can be mistaken for depression or early neurodegenerative disease. Astute clinicians know that through careful history, physical, and appropriate test examinations, the dangerous can be differentiated from the benign.

Case presentation: We report the case a 75-year-old woman who presented after a transient episode of syncope with brief tremulous movements and seizure-like activity. Her history included sudden, unpredictable waves of sadness and intermittent memory lapses, rather than persistent low mood or steadily progressive cognitive decline. Initial evaluation, including imaging and cardiovascular studies, was unremarkable, and her presentation was initially attributed to medication-related bradycardia. Neurological testing, however, revealed three seizure events on spot Electroencephalogram (EEG), fourteen seizures on continuous EEG over 24 h, and associated structural abnormalities on Magnetic Resonance Imaging (MRI) localizing to the left temporal lobe.

Discussion: This case highlights how episodic affective and cognitive symptoms can closely mimic or co-exist with primary psychiatric or neurodegenerative disorders. Recognition of these atypical patterns is critical as untreated seizures can cause significant morbidity. Timely identification and management may also substantially improve overall functioning and quality of life, while avoiding unnecessary treatments and their adverse side effects.

Conclusions: Clinicians should maintain a high index of suspicion for transient neurological events in older adults presenting with atypical mood changes, fluctuating memory, or subtle automatisms.

Keywords: *temporal lobe epilepsy, convulsive syncope, mesial temporal sclerosis*

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1. Introduction

Older adults frequently present with depression and progressive memory changes [1, 2], often leading clinicians to suspect major depressive disorder or early dementia. Yet, temporal lobe epilepsy (TLE) can manifest with affective auras, episodic confusion, and automatisms, closely resembling mood disorders or cognitive decline [3]. The association between temporal lobe seizures and psychiatric phenomena has been recognized since the pioneering work of John Hughlings Jackson in the 1870s, who described the “dreamy state,” a constellation of experiential phenomena including affective symptoms, déjà vu, and altered consciousness [4, 5]. Early clinicopathologic observations, including reports of patients exhibiting complex behaviors under altered consciousness with lesions involving the parahippocampal and uncinate regions, further supported the link between temporal lobe pathology and behavioral and psychiatric manifestations [6]. Psychiatric and behavioral manifestations of temporal lobe seizures have also been described in medico-legal contexts, including cases in which ictal or postictal states were considered in criminal defenses, underscoring the profound behavioral effects of limbic network dysfunction [7].

Moreover, temporal lobe seizures specifically associated with mesial temporal sclerosis (MTS) are frequently accompanied by major depressive disorder and cognitive deficits [8, 9]. Distinguishing these entities is crucial, as epilepsy represents a potentially reversible cause of morbidity. Herein, we describe a patient whose presumed depression and Alzheimer’s-related cognitive impairment were ultimately reinterpreted as manifestations of temporal lobe epilepsy.

2. Case presentation

A 75-year-old female presented to the emergency department after an episode of loss of consciousness, preceded by brief tremulousness and followed by seizure-like activity lasting approximately 30 s. She immediately returned to baseline following the event. Earlier that day, she had fallen out of bed, describing feeling “very scared,” but was unable to provide further details. Her past medical history included depression, treated with escitalopram, and

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mild cognitive impairment attributed to suspected Alzheimer's disease, for which she was prescribed donepezil. Notably, there was no prior history of epilepsy or seizures, and no prior Electroencephalogram (EEG) had been performed.

On arrival to the Emergency Department (ED), her heart rate was 49 in sinus rhythm. A Computerized Tomography (CT) head and bilateral carotid ultrasound were unremarkable. Given her new onset symptomatic sinus bradycardia, her presentation was initially attributed to medication effects, specifically donepezil- or escitalopram-induced bradycardia, and she was diagnosed with probable drug-induced bradyarrhythmic vagovagal syncope or neurally mediated syncope, with a reflex-mediated transient loss of consciousness characterized by sudden vasodilation and/or bradycardia leading to transient cerebral hypoperfusion [10]. However, cardiogenic syncope was also considered.

Given the convulsive component of her syncope, an EEG was obtained to exclude seizure activity. A routine 30 min EEG captured three focal seizures originating from the left temporal lobe, along with mild diffuse slowing (a sample seizure is included in

Figure 1). Continuous EEG monitoring over 24 h subsequently demonstrated 14 focal seizures, again from the left temporal region. MRI brain without gadolinium contrast and with gadolinium contrast was performed and revealed hyperintensity on T2 Fluid-Attenuated Inversion Recovery (FLAIR) imaging in the left mesial temporal lobe indicating potential left MTS (**Figure 2**).

At baseline, she was established at an Alzheimer's institute with a diagnosis of mild cognitive impairment and had a recent Mini-Mental Status Exam (MMSE) of 25 and Montreal Cognitive Assessment (MoCA) of 15. Despite her diagnosis, she was largely independent in activities of daily living. On interviewing the family, they described her memory impairment as sudden episodes of confusion where she would ask the same question repeatedly, rather than a progressive, consistent decline. They additionally characterized her depression as sudden, unpredictable waves of sadness rather than persistent low mood, raising suspicion for atypical affective symptoms. After further interviewing, a family member reported subtle automatisms (e.g., nose wiping, picking at clothes), particularly on the patient's left side, which they had not recognized as clinically significant until questioned.

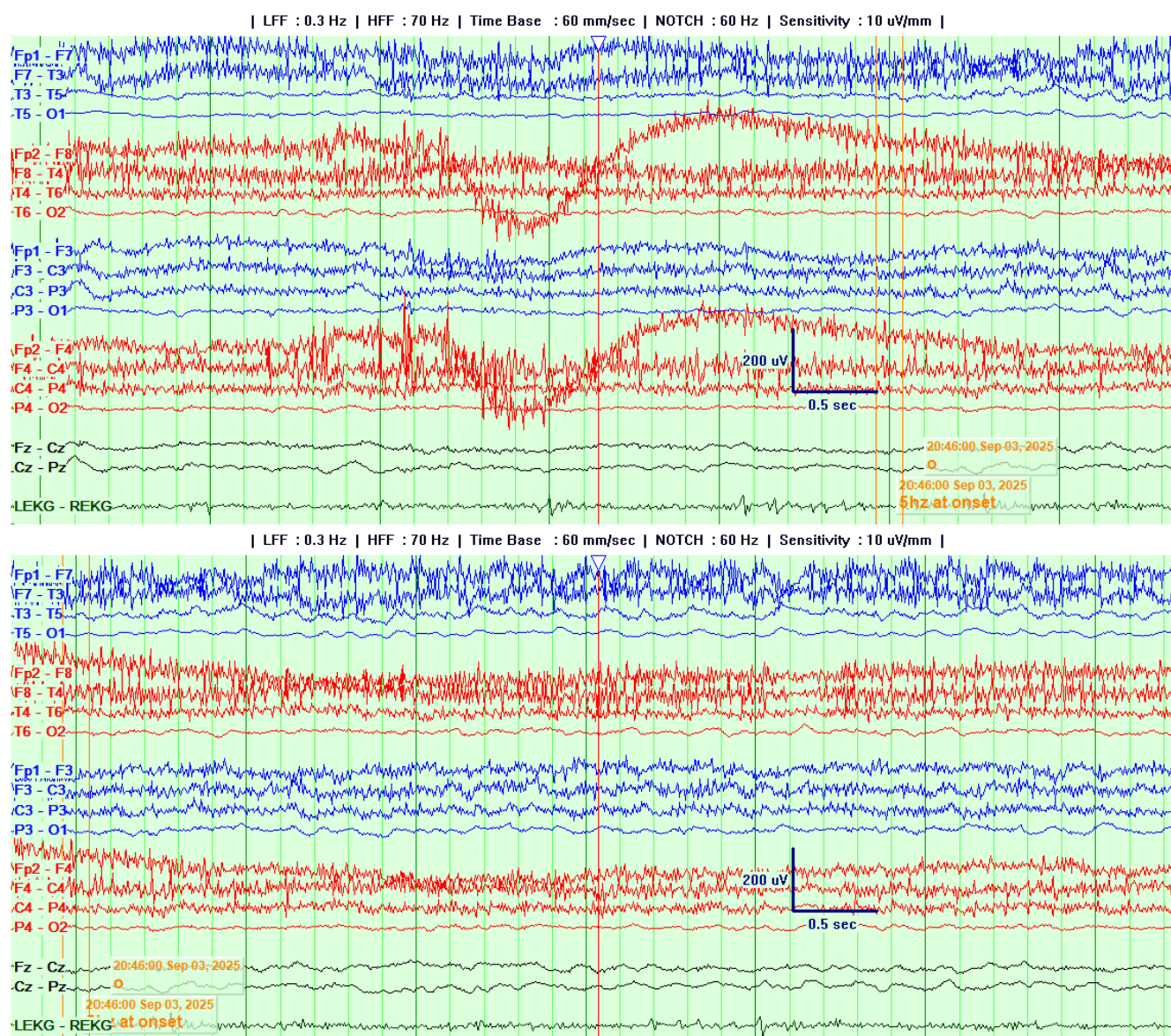


Figure 1 • Cont.

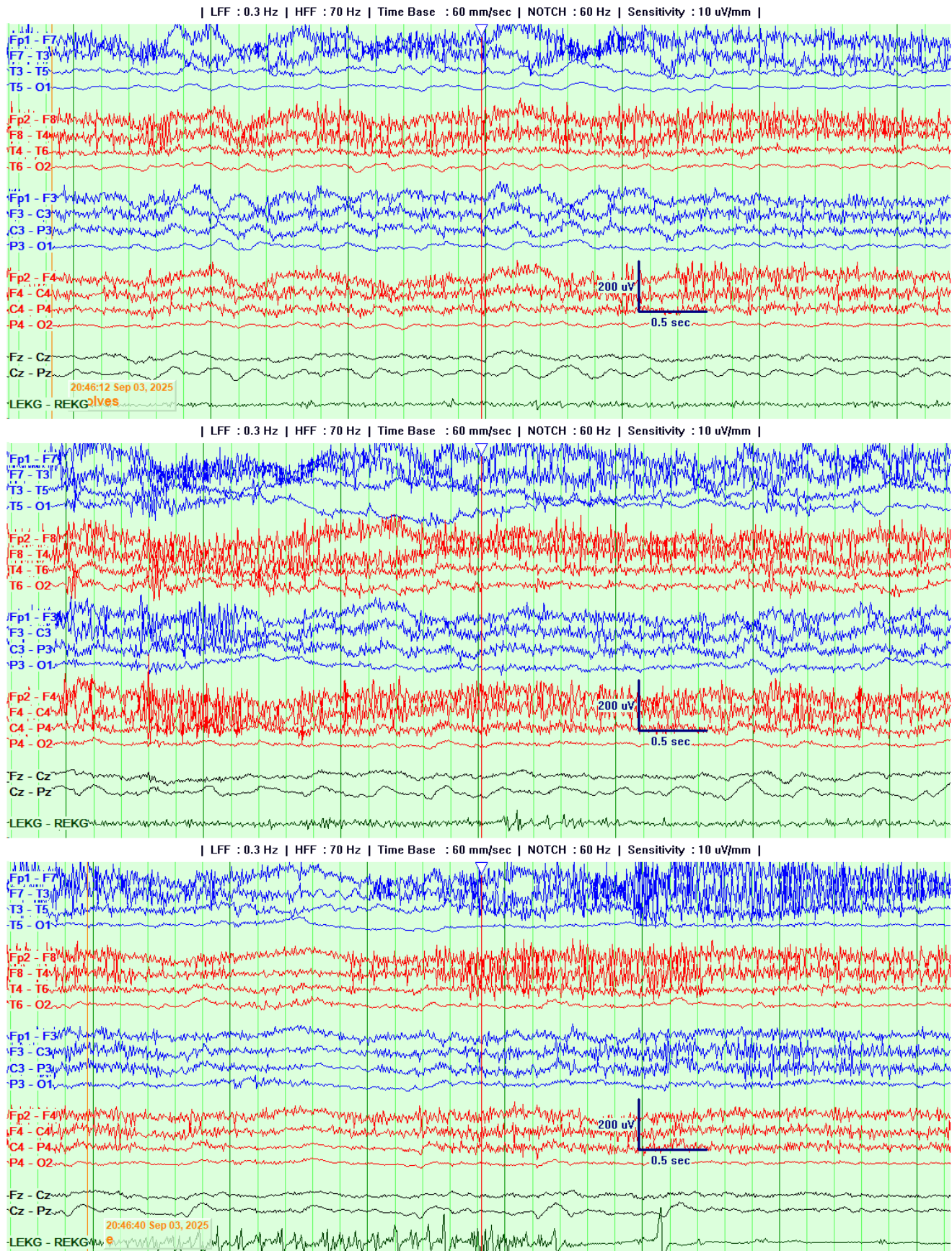


Figure 1 • Initial Electroencephalogram (EEG) with seizure onset in the left temporal region with lower voltage 5 Hz semi-rhythmic theta at F7, T3 and T5 with spread to C3 that evolves into relatively higher voltage activity with sharper and more rhythmic features and slows down to 2 Hz prior to sudden offset. “O” was utilized as a notation for onset of the seizure and “E” was utilized as a notation for the end of the seizure.

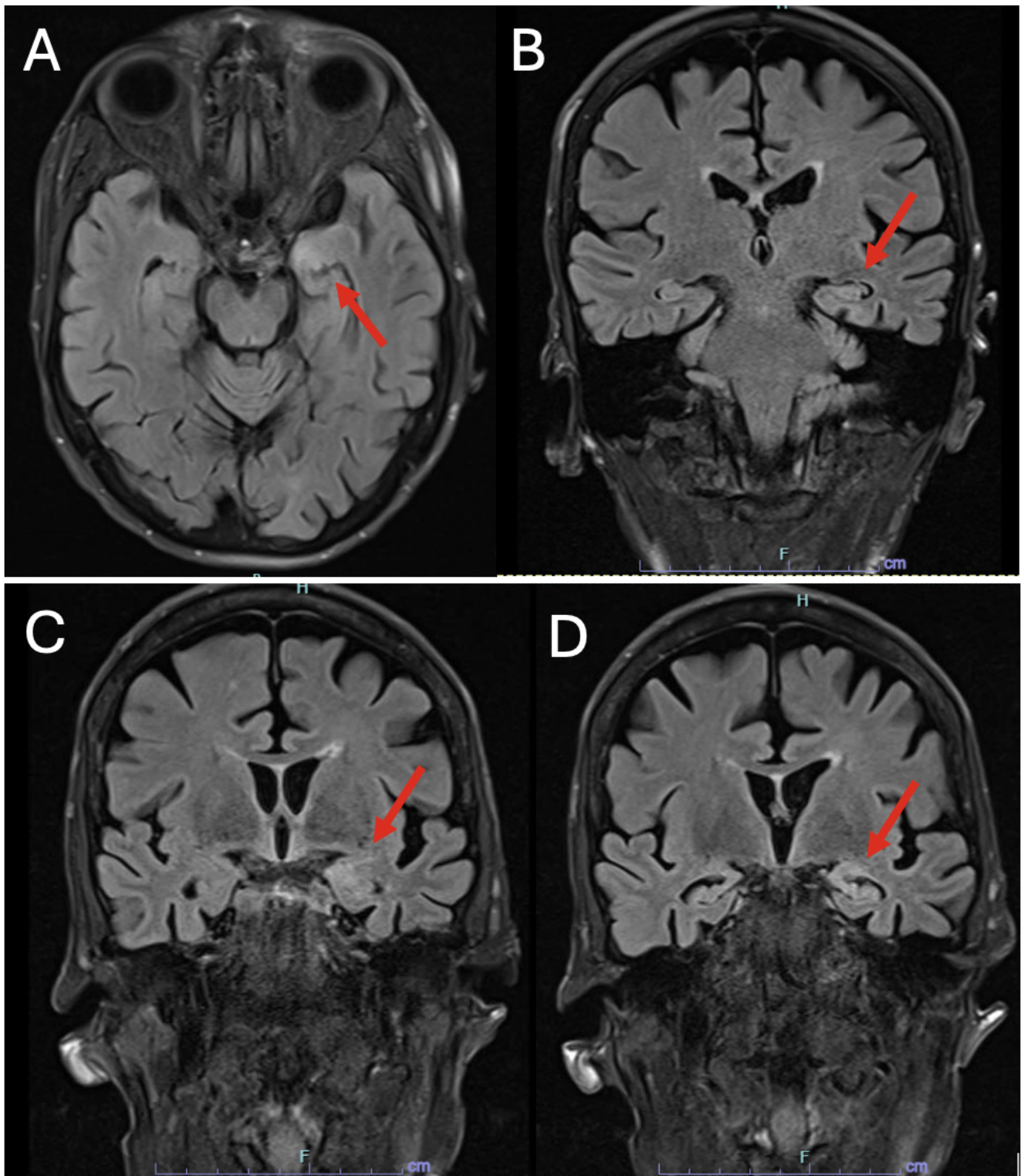


Figure 2 • Magnetic resonance imaging findings of a subtle asymmetric abnormality (indicated by red arrow) suggesting left mesial temporal lobe sclerosis. (A) T2 FLAIR axial image, (B–D) coronal T2 FLAIR images posterior to anterior slices (B–D).

The identification of recurrent focal seizures on EEG, coupled with Magnetic Resonance Imaging of the Brain (MRI) findings of left MTS, suggested that these episodic cognitive and affective disturbances were likely manifestations of seizure activity. This reframing indicated that her fluctuating memory lapses, mood changes, and automatisms were treatable ictal and interictal phenomena, rather than primary psychiatric or irreversible neurodegenerative disease, the treatment of which had precipitated the convulsive syncope leading to her initial presentation.

After admission, escitalopram and donepezil were discontinued, and the patient was initiated on valproate and, due to further seizures, brivaracetam for seizure management with a working diagnosis of temporal lobe epilepsy (TLE) with MTS. Persistent seizure activity on Continuous Electroencephalogram (cEEG) prompted the addition of clobazam. Though her seizures resolved, she subsequently developed hyperammonemia, necessitating the discontinuation of valproate. Her hospital course was further co-

mplicated by transient encephalopathy, likely multifactorial due to hyperammonemia, a known adverse effect of valproate therapy [11]; a urinary tract infection, a common precipitant of delirium in older adults [12]; and sedating effects of antiseizure medications, reflecting the increased vulnerability of older patients to polypharmacy-related cognitive impairment [13]. With targeted treatment of these underlying issues and adjustment of her medication dosage, her condition improved, and she was seizure-free at discharge on dual antiepileptic therapy with brivaracetam 100 mg dosed twice a day and clobazam 5 mg dosed once a day.

Following treatment and medication adjustment, no further clinical seizure-like events were observed during hospitalization. Repeat EEG was obtained (**Figure 3**) which demonstrated generalized slowing and more focal slowing in the left temporal leads. Repeat cognitive testing has not been performed, but subjectively the family feels that the patient's cognition has improved since her initial discharge from the hospital and she is now able to perform basic activities of daily living with family support, as she was prior to the onset of these symptoms. Patient continues to follow up with cardiology, neurology, and her PCP for management of her symptoms at this time.

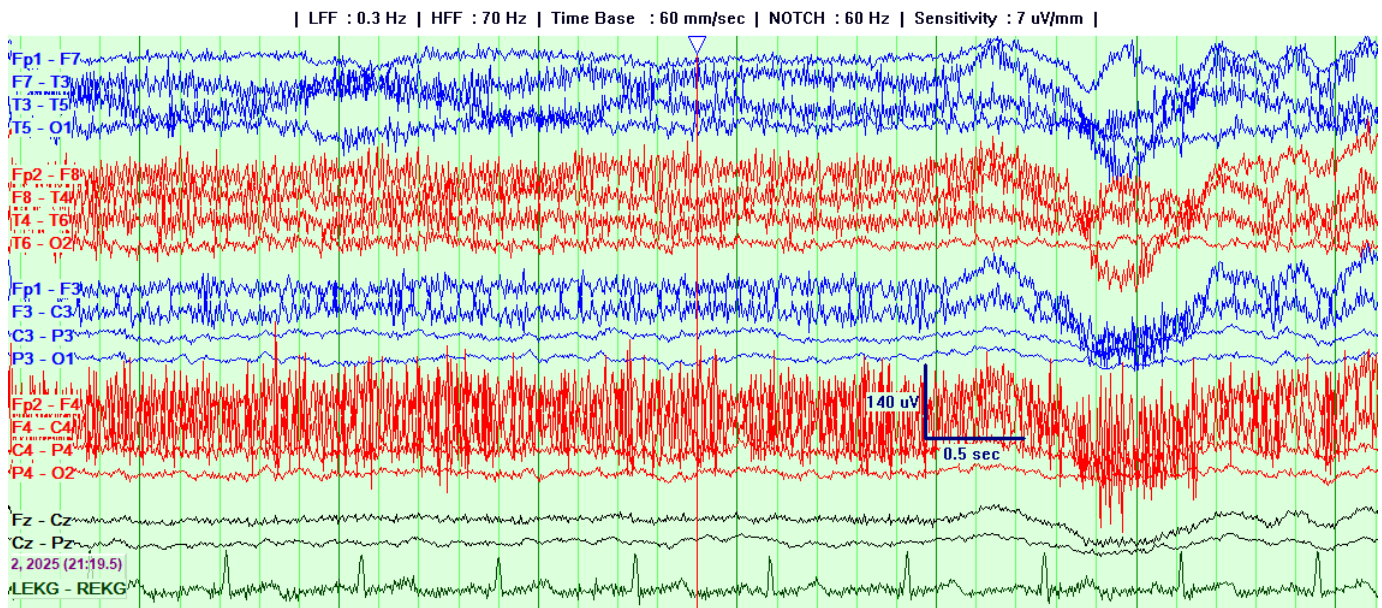


Figure 3 • Sample of repeat EEG with focal slowing seen in the left temporal leads; patient was on extended monitoring for two days without seizure activity.

3. Discussion

This case illustrates how TLE, particularly when associated with MTS, can masquerade as primary psychiatric and cognitive disorders. Our patient was initially diagnosed with depression and mild cognitive impairment due to likely Alzheimer's disease, yet her episodic mood changes and memory lapses were more consistent with ictal and interictal manifestations of TLE. Her sudden "waves of sadness" are typical of affective symptoms seen in TLE, often linked to limbic network dysfunction [3]. Similarly, her memory impairment was not global or steadily progressive but episodic and fluctuating, suggestive of seizure-related cognitive dysfunction rather than primary neurodegeneration.

TLE with MTS is strongly associated with psychiatric comorbidities, including depression and cognitive impairment [3]. These symptoms are not simply psychological reactions to living with a chronic illness but are believed to reflect intrinsic limbic system dysfunction [3]. Studies have shown that patients with TLE, especially those with hippocampal sclerosis, have higher rates of major depressive disorder and cognitive deficits compared to other forms of epilepsy and chronic medical conditions, supporting a direct neurobiological relationship [8, 9, 14–16].

Cognitive impairment in TLE is multifactorial, with hippocampal sclerosis as the primary driver. Deficits most often involve

memory, executive function, and language, and are more pronounced in patients with mesial involvement and left-sided lesions, as in our patient [7, 8]. Depression is also highly prevalent in this population, with rates reported as high as 55% [16], and is associated with both structural and functional abnormalities in the hippocampus and amygdala [12, 15, 17]. The risk is further increased with longer epilepsy duration, greater seizure frequency, and polytherapy [17]. Importantly, psychiatric and cognitive symptoms in TLE are frequently misdiagnosed as primary mood or neurodegenerative disorders, delaying appropriate seizure management [3]. Early recognition and treatment of seizures can lead to significant improvements in both mood and cognitive outcomes [3, 16].

Importantly, epilepsy and neurodegenerative disease are not mutually exclusive, with growing evidence supporting bidirectional associations between epilepsy and dementia, including Alzheimer's disease [18]. Epilepsy in older adults frequently occurs in the context of an underlying neurodegenerative pathology [19]. The mild diffuse slowing observed on EEG in this patient is nonspecific and may reflect medication effects, metabolic factors, or early neurodegeneration, and does not exclude a coexisting Alzheimer's pathology [20]. This case therefore supports a model of dual pathology, in which epileptic activity represents a superimposed and potentially modifiable contributor to cognitive and affective dysfunction [21].

Several lines of evidence support a clinically meaningful epileptic contribution. The patient's symptoms were episodic and fluctuating, continuous EEG demonstrated a high seizure burden, and semiology was consistent with temporal lobe seizures [22]. MRI findings of left mesial temporal sclerosis provided a structural correlate [23]. While causality cannot be definitively established in a single case, the convergence of clinical history, EEG findings, and imaging supports a significant contribution from seizure activity. The observed clinical improvement following treatment further supports this interpretation, although the absence of formal follow-up testing limits definitive conclusions [19].

This case also highlights the risks of iatrogenic complications in older adults. The patient developed hyperammonemia and encephalopathy related to antiseizure therapy, underscoring the need for careful medication selection and monitoring in this population [11, 13].

4. Conclusions

This case demonstrates how temporal lobe epilepsy can mimic or co-exist with both depression and dementia, leading to diagnostic delay. Elderly patients presenting with sudden mood shifts, affective auras, or episodic memory loss should be evaluated for seizures, especially when conventional treatments for depression or cognitive decline do not fully explain or improve symptoms. The timely recognition of epilepsy as an underlying cause offers the potential for effective intervention and improved patient outcomes.

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Author contributions

Conceptualization, T.V. and C.C.; methodology, J.K. and T.V.; validation, T.V. and C.C.; formal analysis, J.K. and T.V.; investigation, J.K., T.V. and A.M.; resources, C.C.; data curation, J.K. and T.V.; writing—original draft preparation, J.K. and T.V.; writing—review and editing, J.K., T.V., A.M., J.C. and C.C.; visualization, J.K.; supervision, T.V. and C.C.; project administration, T.V. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

Christina Chrisman: None relevant to this article; Alexion speaker bureau myasthenia gravis, argenx speaker bureau myasthenia gravis and CIDP, UCB speaker bureau myasthenia gravis, Grifols speaker bureau CIDP, Biogen advisory board SMA, and CSL advisory board CIDP. The remaining authors declare that they have no competing interests.

Data availability statement

All data supporting the findings of this publication are available within this article.

Institutional review board statement

Ethical review and approval were waived for this study, due to the nature of this work as a single-patient case report. The patient provided written informed consent for publication of her de-identified clinical information and imaging. Case reports involving a single patient with informed consent are exempt from formal IRB review according to institutional policy, consistent with the Common Rule (45 CFR 46) which does not classify case reports as human subjects research requiring IRB oversight.

Informed consent statement

Written informed consent has been obtained from the participant to publish this paper.

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